



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
01-08-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TIAS, S, I(A), I, PHASE LIVE-I & I(A)]

Time : 3 Hours

PAPER-2

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

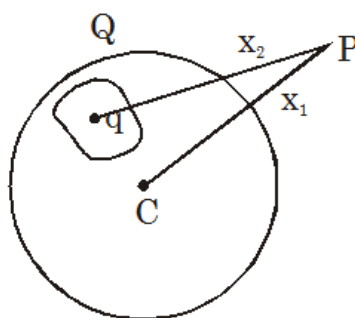
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A solid conducting sphere is given charge Q and has a charge q inside the cavity as shown. Choose correct option(s) regarding electric field & potential at point P.



(A) Electric field at point P is $\frac{kQ}{x_1^2}$

(B) Electric field at point P due to q is $\frac{kq}{x_2^2}$

(C) Potential at point P is $\frac{kQ}{x_1} + \frac{kq}{x_2}$

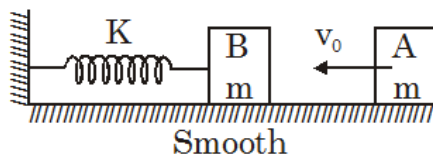
(D) Potential at point P is $\frac{kQ}{x_1} + \frac{kq}{x_1}$

Space for Rough Work

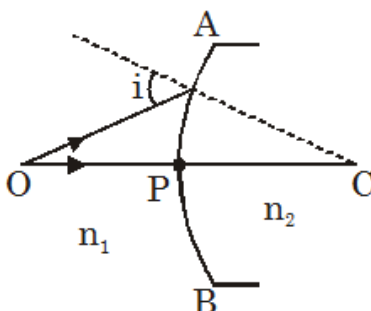
2. A gas is found to obey the law $P^2V = \text{constant}$. The initial temperature and volume are T_0 and V_0 . If the gas expands to a volume $3V_0$,
- (A) final temperature becomes $\sqrt{3}T_0$ (B) Internal energy of the gas will increase
- (C) Final temperature becomes $\frac{T_0}{\sqrt{3}}$ (D) Internal energy of the gas decreases.
3. During the melting of a slab of ice at 273 K at atmosphere pressure:
- (A) Positive work is done by the ice–water system on the atmosphere.
- (B) Positive work is done on the ice water system by the atmosphere.
- (C) The internal energy of ice–water system increases.
- (D) The internal energy of ice–water system decreases.
4. In a mixture of nitrogen and helium kept at room temperature, as compared to a helium molecule, nitrogen molecule hits the wall
- (A) with greater average speed (B) with smaller average speed
- (C) with greater average kinetic energy (D) with smaller average kinetic energy.

Space for Rough Work

5. A block A of mass m is given a velocity v_0 , towards another block B of same mass. B is attached to an ideal spring of spring constant K . A makes a head on perfectly inelastic collision with B. Choose the correct option(s).



- (A) Time after the collision, B attains its initial position again for the second time is $\pi\sqrt{\frac{m}{K}}(\sqrt{2} + 1)$
- (B) Time after the collision, B attains its initial position again for the second time is $\pi\sqrt{\frac{2m}{K}}$
- (C) Maximum elongation in the spring during subsequent motion is $\frac{1}{2}\sqrt{\frac{mv_0^2}{K}}$
- (D) Maximum elongation in the spring during subsequent motion is $\sqrt{\frac{mv_0^2}{2K}}$
6. A point object O is kept at a distance of $OP = u$. The radius of curvature of the spherical surface APB is $CP = R$. The refractive index of the media are n_1 and n_2 which are as shown in the diagram. Then, (consider only paraxial rays)



- (A) If $n_1 > n_2$, image is virtual for all values of 'u'.
- (B) If $n_2 = 2n_1$, image is virtual when $R > u$.
- (C) The image is real for all values of u , n_1 and n_2 .
- (D) If $n_2 > n_1$, then image will be always real

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

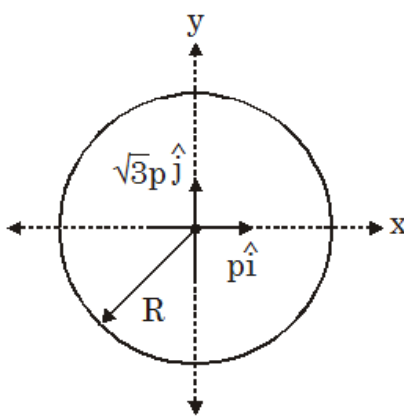
- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. List-I gives a situation in which two dipoles of dipole moment $p \hat{i}$ and $\sqrt{3}p \hat{j}$ are placed at origin. A circle of radius R with centre at origin is drawn as shown in figure. List-II gives coordinates of certain positions on the circle. Match the statement(s) in list-I with the statement(s) in list-II.



	List-I		List-II
(P)	The coordinate(s) of point on circle where potential is maximum	(1)	$\left(\frac{R}{2}, \frac{\sqrt{3}R}{2}\right)$
(Q)	The coordinate(s) of point on circle where potential is zero.	(2)	$\left(-\frac{R}{2}, -\frac{\sqrt{3}R}{2}\right)$
(R)	The coordinate(s) of point on circle where magnitude of electric field intensity is $\frac{1}{4\pi\epsilon_0} \frac{4p}{R^3}$	(3)	$\left(-\frac{\sqrt{3}R}{2}, \frac{R}{2}\right)$
(S)	The coordinate(s) of point on circle where magnitude of electric field intensity is $\frac{1}{4\pi\epsilon_0} \frac{2p}{R^3}$	(4)	$\left(\frac{\sqrt{3}R}{2}, -\frac{R}{2}\right)$

(A) $P \rightarrow 1; Q \rightarrow 3, 4; R \rightarrow 1, 2; S \rightarrow 3, 4$

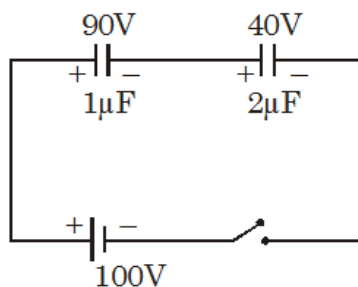
(B) $P \rightarrow 2; Q \rightarrow 3, 4; R \rightarrow 1; S \rightarrow 1, 2$

(C) $P \rightarrow 3; Q \rightarrow 1, 2; R \rightarrow 3, 4; S \rightarrow 3, 4$

(D) $P \rightarrow 1; Q \rightarrow 1, 2; R \rightarrow 2, 3; S \rightarrow 3, 4$

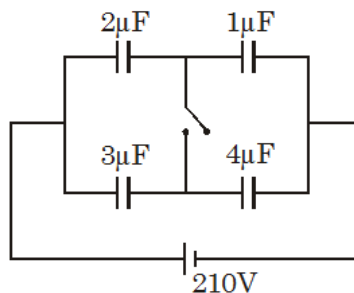
Space for Rough Work

8. Initially charged two capacitors and a battery of 100 V are arranged as shown in the figure. Now the switch S is closed.



	List-I		List-II
(P)	Final charge on $1\mu\text{F}$ capacitor	(1)	20
(Q)	Final charge on $2\mu\text{F}$ capacitor	(2)	2000
(R)	Charge flown through switch	(3)	60
(S)	Total work done by battery during the process (magnitude)	(4)	70

- (A) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$ (B) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$
 (C) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$ (D) $P \rightarrow 4; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 3$
9. Consider a network shown in the figure. Initially the switch S was open. If the switch S is closed now

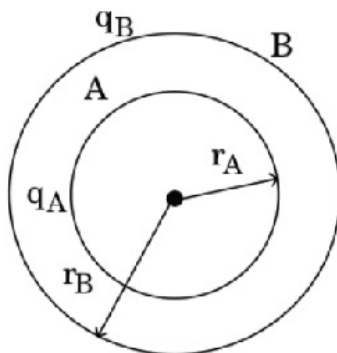


	List-I		List-II
(P)	Initially charge on $2\mu\text{F}$ capacitor	(1)	0.210×10^{-3}
(Q)	Finally charge on $2\mu\text{F}$ capacitor	(2)	0.140×10^{-3}
(R)	Charge flown through the switch	(3)	2.625×10^{-3}
(S)	Heat loss	(4)	0.105×10^{-3}

- (A) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$ (B) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$
 (C) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$ (D) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

Space for Rough Work

10. A charged spherical conductor A of radius r_A is surrounded by a hollow conductor B of radius r_B such that $r_B > r_A$ as shown in Figure. A is given a charge q_A and that of B, q_B . V_A and V_B are the potentials of A and B respectively. Now match the items in list-I with those in list-II for the conditions indicated in list-I.



	List-I		List-II
(P)	$q_A = +mC$ and $q_B = +nC$ and $m < n$	(1)	potential difference depends on q_A only
(Q)	$q_A = -mC$ and $q_B = -nC$ and $m > n$	(2)	$V_A > V_B$
(R)	$q_A = -mC$ and $q_B = -nC$ and $m < n$ and A and B are connected by a conducting wire	(3)	$V_A < V_B$
(S)	$q_A = -mC$ and $q_B = -nC$ and $m > n$ and A and B are connected by a conducting wire	(4)	charge flows from A to B
		(5)	$q_A = 0$

- (A) $P \rightarrow 1,2; Q \rightarrow 1,3; R \rightarrow 1,4,5; S \rightarrow 1,4,5$
 (B) $P \rightarrow 1,4; Q \rightarrow 2,3; R \rightarrow 2,4; S \rightarrow 4,5$
 (C) $P \rightarrow 1,3; Q \rightarrow 3,4; R \rightarrow 1,4,5; S \rightarrow 1,4$
 (D) $P \rightarrow 2,4; Q \rightarrow 1,3; R \rightarrow 1,2,3; S \rightarrow 1,3$

Space for Rough Work

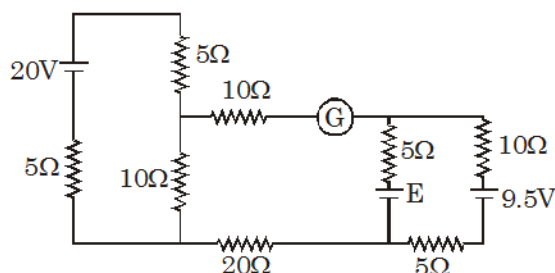
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

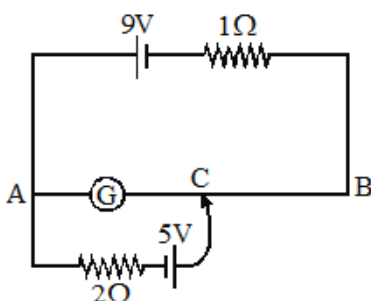
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. What should be value of E (in volt) for which galvanometer shows no deflection :

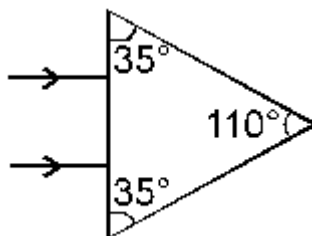


2. Consider a cube of side $a = 0.1$ m placed such that its six faces are given by equations $x = 0$, $x = +a$, $y = 0$, $y = +a$, $z = 0$ and $z = +a$, placed in electric field given by $\vec{E} = x^2\hat{i} + y\hat{j}$ N/C. Find the electric flux crossing out of the cube in the unit of 10^{-4} N m²/C.
3. The resistance of uniform wire AB is 9Ω . Find the resistance of part BC (in Ω) for which galvanometer G will show zero deflection.



Space for Rough Work

4. A conducting soap bubble of average radius R and thickness t is charged to a potential V . The bubble breaks into two identical spherical drops. The potential of any of the drops is given by $V' = \frac{R}{((3.2) n R^2 t)^{\frac{1}{3}}} V$, where n is an integer. What is the value of n ?
5. Two parallel beams of light pass through an isosceles prism of refractive index 1.4 as shown in the figure. The angle between the two emergent rays is (in degree). [$\sin 35^\circ = 0.57$, $\sin^{-1}(0.798) = 53^\circ$]



6. The radii of a spherical capacitor electrodes are equal to a and b ($a < b$). The space between them is filled with a dielectric of dielectric constant K and resistivity ρ . Initially at time $t = 0$, the inner electrode gets a charge q_0 . The amount of heat generated during the spreading of charge is $H = \left(\frac{1}{a} - \frac{1}{b} \right) \frac{q_0^2}{2.5x\pi\epsilon_0 K}$, the value of x is :

Space for Rough Work

7. A very long uniformly charged thread oriented along the axis of a circle of radius $R = 1\text{m}$, rests on its centre with one of the ends. The charge per unit length on the thread is $\lambda = 19\epsilon_0$. Find the flux of the vector E through the circle area in (Vm).
8. The electric field in a region is radially outward with magnitude $E = 2r$. The charge contained in a sphere of radius $a = \frac{5}{2}\text{m}$ centred at the origin is $4x\pi\epsilon_0$. Find the value of x .
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Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. In which of the following electronic configuration, Pauli exclusion principle is violated?

(A) $1s^2 2s^2 2p^6 3s^1 3p^3 3d^2$

(B) $1s^1 2s^1 2p^3 3s^3$

(C) $1s^2 2s^2 2p^7$

(D) $\begin{array}{|c|c|c|c|} \hline \uparrow\downarrow & \uparrow\downarrow & \uparrow\downarrow & \uparrow\downarrow \\ \hline 1s & 2s & 2p & \end{array}$
2. When a sulphide ore of a noble metal is treated with excess of cyanide solution then a soluble complex is formed which when treated with Zn, a soluble complex (X) is formed and noble metal is obtained as residue. Which of the following statement is correct regarding (X)

(A) Zn is present in anionic part

(B) Hybridisation is dsp^2

(C) EAN value is 36

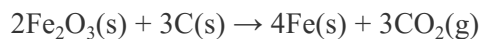
(D) show synergic bonding

Space for Rough Work

3. Select the correct option(s) according to second law of thermodynamics.
- (A) $(\partial S)_{U,V} \geq 0$
 - (B) $(\partial H)_{P,S} \geq 0$
 - (C) $(\partial G)_{P,T} \leq 0$
 - (D) $(\partial U)_{S,V} \leq 0$
4. For which of the following process entropy of system containing ideal gas increases
- (A) Single stage adiabatic compression
 - (B) Reversible adiabatic expansion
 - (C) Irreversible adiabatic expansion
 - (D) Isobaric cooling
-

Space for Rough Work

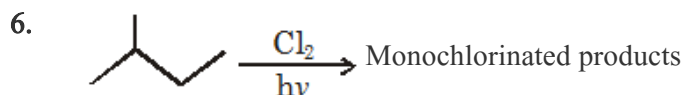
5. Consider the following reaction :



ΔH_f° of Fe_2O_3 & CO_2 are -820 kJ/mol and -390 kJ/mol respectively.

The reaction is -

- (A) Endothermic
- (B) Exothermic
- (C) Spontaneous at high temperature
- (D) Spontaneous at low temperature



Correct statement regarding above reaction :

- (A) Total 5 monochlorinated product will form
- (B) Total 4 fraction are obtained after fractional distillation product mixture
- (C) Total 4 chiral products will form
- (D) 3 enantiomeric pairs are formed

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. Answer the following by appropriately matching the lists based on the information given in the list.

	List-I		List-II
(P)	N_2^+ & NO	(1)	Same bond order as that of B_2
(Q)	NF^{-2} & Cl_2	(2)	Same magnetic property and bond order as that of NO^+
(R)	CO & CN^-	(3)	Same number of unpaired electron as that of S_2^-
(S)	C_2 & CN^+	(4)	Same bond order but not same magnetic property as that of O_2
		(5)	The bond order is less than that observed in CO^+
		(6)	Similar magnetic property as that of O_3

Correctly match among the following is :

- (A) $\text{P} \rightarrow 3,4,5; \text{Q} \rightarrow 1,4,6; \text{R} \rightarrow 5; \text{S} \rightarrow 4,5,6$
 (B) $\text{P} \rightarrow 3,5,6; \text{Q} \rightarrow 1,4,5; \text{R} \rightarrow 2; \text{S} \rightarrow 5,6$
 (C) $\text{P} \rightarrow 3,5; \text{Q} \rightarrow 1,5,6; \text{R} \rightarrow 2,5,6; \text{S} \rightarrow 4,5,6$
 (D) $\text{P} \rightarrow 3,4,5; \text{Q} \rightarrow 1,5; \text{R} \rightarrow 2,4,5; \text{S} \rightarrow 4,6$

Space for Rough Work

8. For process $\text{H}_2\text{O}(l) (100^\circ\text{C}, 1 \text{ atm}) \rightleftharpoons \text{H}_2\text{O}(g) (100^\circ\text{C}, 1 \text{ atm})$

List-I		List-II	
(P)	ΔU	(1)	0
(Q)	ΔS (system + surrounding)	(2)	ΔH
(R)	Work	(3)	Positive
(S)	Heat involved	(4)	Negative

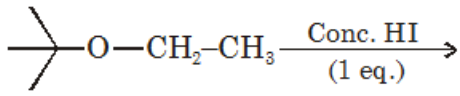
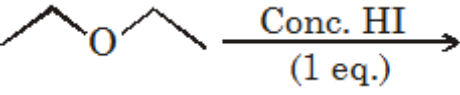
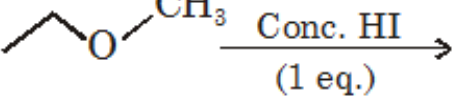
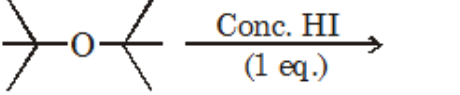
(A) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 2$

(B) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$

(C) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

(D) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$

9.

	List-I (Reaction)		List-II (Product(s) corresponding to reactions)
(P)		(1)	
(Q)		(2)	
(R)		(3)	CH_3I
(S)		(4)	

The correct option is :

(A) $P \rightarrow 1,4; Q \rightarrow 2,4; R \rightarrow 3,4; S \rightarrow 1$

(B) $P \rightarrow 1,2; Q \rightarrow 2,3; R \rightarrow 1,4; S \rightarrow 1$

(C) $P \rightarrow 1,2; Q \rightarrow 2,3; R \rightarrow 2,4; S \rightarrow 4$

(D) $P \rightarrow 3,4; Q \rightarrow 2,3; R \rightarrow 1,2; S \rightarrow 2,3$

Space for Rough Work

10.

	List-I (Reaction)		List-II (Product)
(P)		(1)	Pair of enantiomers
(Q)		(2)	Pair of diastereomers
(R)		(3)	Meso product
(S)		(4)	Optically pure product

The correct option is :

(A) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 3$

(B) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$

(C) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 3$

(D) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

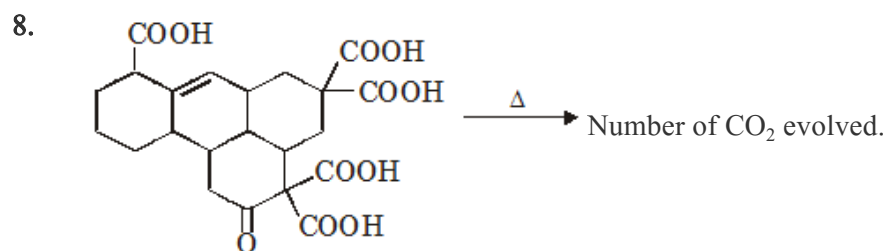
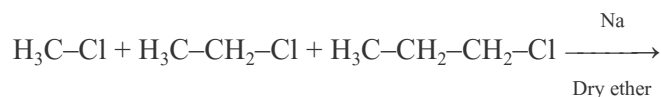
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Number of ores which produce CO_2 on heating in absence of oxygen in reverberatory furnace.
Pyrolusite, Sphalerite, Azurite, Cerrusite, Siderite, Ilmenite, Magnesite, Calamine, Anglesite
2. The change in oxidation state of Pb when galena is roasted for the lead extraction from it's low grade ore.
3. For an octahedral complex of Fe^{3+} the value of C.F.S.E. (Crystal field stabilization energy) is " $x\Delta_0$ " if magnetic moment of complex is $\sqrt{35}$ B.M. Find out the value of $|x|$. (Δ_0 is crystal field splitting energy in octahedral field)
4. Find maximum number of electrons in Al for which $\frac{\ell \times m}{n} = 0$ is

Space for Rough Work

5. For the reaction $A(g) \rightleftharpoons B(g)$; $K_p = 10$ at 350K. Calculate ΔG (in kcal/mole) at 350 K, if partial pressures of A & B gases are 10 bar & 1 bar respectively. $\left[\text{Given : } R = 2 \frac{\text{cal}}{\text{mol} \cdot \text{K}} \right]$
6. The half life of a radionuclide X^{200} is 14×10^5 min. If its specific activity is $x \times 10^{13}$ dps per gram, the value of 'x' is ($N_A = 6 \times 10^{23}$, $\ln 2 = 0.7$)
7. Find the total number of different possible combination product (Considering only combination of radical but not disproportionation of radical) for following Wurtz reaction :



Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

- Let $y = f(x)$ is a function such that $f(0) = 0$ and $\frac{dy}{dx} = xe^{y-x^2}$, then
 (A) $f(x)$ is an even function (B) $f(x)$ is an odd function
 (C) $f(x) \geq 0 \forall x \in \mathbb{R}$ (D) $\lim_{x \rightarrow \infty} \frac{f'(x)}{x} = 4$
- If $\int (\cos x) d(\sin x) = f(x) + g(x) + C$ (where $f(x)$ is aperiodic function & $g(x)$ is periodic function) and $f(2) = 1$, $g\left(\frac{\pi}{4}\right) = \frac{1}{4}$, then
 (Where C is constant of integration)
 (A) $f(0) = 0$ (B) $f(0) = 1$ (C) $g(0) = 0$ (D) $4g(0) + 1 = 0$

Space for Rough Work

3. If circles of radius of R and r ($R > r$) touch each other externally such that triangle formed by their common tangents is equilateral triangle of area $9\sqrt{3}$ sq. units, then
- (A) $\frac{R}{r} = 2$ (B) $\frac{R}{r} = 3$
- (C) Length of their direct common tangent = 6 (D) Length of their direct common tangent = $2\sqrt{6}$
4. Let $f(x) = e^x(x^2 + x + 1) \forall x \geq 0$ and $f^{-1}(x)$ be its inverse. Let $I_1 = \int_0^1 f(x) dx$ and $I_2 = \int_1^{3e} f^{-1}(x) dx$, then
- (A) $I_1 + I_2 = 3e$ (B) $I_2 - I_1 = 2e + 1$ (C) $I_1 > I_2$ (D) $I_1 < I_2$
5. If area bounded by curve $y = 2\cos^{-1}x$, x-axis and y-axis divided by curve $y = \sin^{-1}x$ into two parts whose areas are A_1 and A_2 ($A_1 < A_2$), then
- (A) $A_2 = 3A_1$ (B) $A_2 = 2A_1$
- (C) $10A_1$ is a prime number (D) $10A_2$ is a prime number
6. If $y = e^{x + e^{x + e^{x + e^{\dots \infty}}}}$, then at $x = 1$
- (A) $\frac{dy}{dx} \in (0, e^2)$ (B) $y \in (0, 1)$ (C) $y > \frac{dy}{dx}$ (D) $y \leq \frac{dy}{dx}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. Let $f(x) = \lim_{n \rightarrow \infty} \prod_{n=0}^n \frac{(x^{2^n} + 1)^2}{(x^{2^{n+1}} + 1)} \forall x \in [0, \infty) \sim \{1\}$ & $f(1) = 1$, then

	List-I		List-II
(P)	$f(x)$ is increasing in	(1)	$x \in (1, \infty)$
(Q)	$f(x)$ is decreasing in	(2)	$x \in (0, \frac{1}{2}) \cup (5, \infty)$
(R)	$f(x)$ has a local minima in	(3)	$x \in (0, 1)$
(S)	$f(x) = x$ has only one solution in	(4)	$x \in (0, 2)$
		(5)	$x \in (0, 1) \cup (4, 5)$

(A) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 4$

(B) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$

(C) $P \rightarrow 4; Q \rightarrow 5; R \rightarrow 3; S \rightarrow 4$

(D) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 5; S \rightarrow 3$

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8.

	List-I		List-II
(P)	$\int_1^5 \sqrt{x - 2\sqrt{x-1}} dx$	(1)	$\frac{\pi}{2}$
(Q)	$\int_{-1}^2 \left[\frac{[x]}{1+x^2} \right] dx$ where $[.]$ denotes greatest integer function	(2)	2
(R)	$\int_{1/2}^2 \frac{x \ln x}{(1+x^2)^2} dx$	(3)	-1
(S)	$\int_0^\pi \left \sin x - \frac{2x}{\pi} \right dx$	(4)	0
		(5)	$\frac{\pi}{4}$

(A) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$

(B) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 2$

(C) $P \rightarrow 3; Q \rightarrow 5; R \rightarrow 2; S \rightarrow 1$

(D) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$

9. Tangents drawn from origin to the circle $S_1 : x^2 + y^2 - 12x - 16y + c = 0$ ($c > 50$) touching circle S_1 at points A and B and $AB = 5\sqrt{3}$. Circle $S_2 = 0$ whose centre is (h, k) and radius r passes through A & B and orthogonal to S_1 , then

	List-I		List-I
(P)	$\sqrt{c} =$	(1)	2
(Q)	$h + 3 =$	(2)	3
(R)	$k + 4 =$	(3)	4
(S)	$r =$	(4)	$5\sqrt{2}$
		(5)	$5\sqrt{3}$

(A) $P \rightarrow 5; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$

(B) $P \rightarrow 5; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 5$

(C) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 5$

(D) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 5; S \rightarrow 1$

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10.

	List-I		List-II
(P)	Let $f(x)$ be a continuous function in $[-1, 1]$ such that $f(x) = \begin{cases} \frac{e^{x^2} + ax + b}{x^2} & ; -1 \leq x < 0 \\ 1 & ; x = 0 \\ \frac{\tan(\tan x)}{x} & ; x > 0 \end{cases}$, then $(a + b)$ is	(1)	-1
(Q)	If $f(x) = \begin{cases} px^2 + qx & ; x \leq -2 \\ px + 2 & ; x > -2 \end{cases}$ is differentiable for all $x \in \mathbb{R}$ ($p \neq 0, q \neq 0$), then $\left \frac{q}{p} \right $ is	(2)	0
(R)	If $\lim_{x \rightarrow 0} \sqrt{\frac{1 - \cos \frac{x}{t}}{x^2}} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x \cdot \cos \frac{\pi}{4}}{(\frac{\pi}{2} - x)}$, then the value of t is	(3)	2
(S)	Consider a triangle ABC with $AB = AC = r$ whose area is Δ , $\angle BCA = r$ (radian) and a sphere with radius r with volume V , then $\lim_{r \rightarrow 0} \frac{16\pi\Delta}{3V}$ is	(4)	4
		(5)	5

(A) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 4$

(B) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$

(C) $P \rightarrow 1; Q \rightarrow 5; R \rightarrow 1; S \rightarrow 4$

(D) $P \rightarrow 2; Q \rightarrow 5; R \rightarrow 2; S \rightarrow 3$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If 'a' is the number of point of extremum of the function $f(x) = \int_0^{x^2-1} t(t-1)(t-3)^3 dt$ then a^2 is
2. If area bounded by the curve $y^2 = x - y$, $y = -1$ and tangent to curve at origin is Δ then $\frac{1}{\Delta}$ is
3. If $\int (\sin x)^{-\frac{3}{4}} (\cos x)^{-\frac{21}{4}} dx = \frac{(\tan x)^{a_1}}{b_1} + \frac{(\tan x)^{a_2}}{b_2} + \frac{(\tan x)^{a_3}}{b_3} + C$ where C is integration constant, then $\left(\frac{a_1 + a_2 + a_3 + b_1 + b_2 + b_3}{2} \right)$ equals
4. Let PQR is a right angled triangle right angled at Q and PQ = 6, QR = 8. Radius of a circle which touches PR at mid point of PR and passes through mid point of PQ, is

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5. If $f(x)$ is integrable function in $\left[\frac{\pi}{6}, \frac{\pi}{3}\right]$ and $I_1 = \int_{\pi/6}^{\pi/3} \tan^2 \theta f(2 \operatorname{cosec} 2\theta) d\theta$ and $I_2 = \int_{\pi/6}^{\pi/3} \cot^2 \theta f(2 \operatorname{cosec} 2\theta) d\theta$ then $\frac{I_1}{I_2}$ is equal to-
6. $\tan\left(\cos^{-1} \frac{2}{\sqrt{5}} + 2 \tan^{-1}(\sqrt{10} - 3)\right)$ is equal to
7. Consider a function $y = f(x)$ ($f(x) > 0$) satisfying equation $f'(x) = xf(x) \forall x \in \mathbb{R}$ and $y = x$ is a tangent to the $f(x)$, then $\ln(f(4))$ is equal to
8. $\lim_{n \rightarrow \infty} \left(\sqrt{\frac{1}{n^3 - 1}} + \sqrt{\frac{2}{n^3 - 2^3}} + \sqrt{\frac{3}{n^3 - 3^3}} + \dots + \sqrt{\frac{n}{n^3 - (n-1)^3}} \right)$ is equal to

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Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

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